

Section 2.2: Direct Input Devices

In section 2.1 we looked at manual input devices and talked about how these require a large amount of human interaction in order to get data into a computer.

In this section we will focus on direct input devices. Direct input devices require **less human interaction** in order to get data into a computer.

Key Concepts of this section:

- # Know the **purpose of** and be able to **identify** different direct input devices.
- # Understand the **uses of** each direct input device.
- # Be able to **discuss advantages and disadvantages** of each direct input device.

Direct Input Devices (no/little human interaction needed)



Key Words:

Magnetic Stripe Reader, Chip and Pin Reader, Barcode Reader, Magnetic Ink Character Recognition (MICR), Optical Mark Recognition (OMR), Optical Character Reader (OCR)

Examples of Direct Input Devices:

- # The table below shows you some examples of direct input devices. Click each device to move to more information:

Magnetic Stripe Reader



Chip and PIN Reader



Barcode Reader



Optical Mark Reader (OMR)

Optical Character Reader (OCR)

Magnetic Ink Character Recognition (MICR)

REMEMBER!

Direct input devices do not require much human interaction to get their data into a computer system.

For example:

All a human has to do to use a Barcode Reader is point the reader at the barcode and pull the trigger. The reader will do the rest.

NOTE:

Click the images to the left to move to more information about each one.



Breakdown of each direct input device:

The following sections will help summarise each of the direct input devices. For each one you will see:

- **Information** about the device
- **Uses** of the device
- **Advantages** of the device
- **Disadvantages** of the device

1 Magnetic Stripe Reader

Information:

- # Reads information from **magnetic stripes** found on **Bank Cards, Membership Cards** and **Hotel Door Cards** etc.
- # The strip on the cards holds information such as:
 - **Bank account number**
 - **Name of card holder**
 - **Expiry date of membership**
 - **etc**
- # Data contained on the card's stripe is read by pulling the card through the magnetic stripe reader.

This is known as '**swiping**'.

- # The magnetic stripe reader then sends data from the card into the computer for processing.

Uses of a Magnetic Stripe Reader:

- # **ATM's** use these readers to process the information on **bank cards**.
- # **EFTPOS** (Electronic Funds Transfer Point of Sale) use the readers to **transfer customer's money** from their bank accounts when they purchase goods in stores.
- # Hotel rooms sometimes use magnetic stripe readers **in place of door keys**.

People who want to enter a room would swipe their card through the reader in order to open the door.

Examples:



Information held on a card's 'magnetic stripe' is read by swiping the card through a strip reader.



Details about bank accounts are stored on a magnetic stripe .

Advantages/Disadvantages of using a Magnetic Stripe Reader:

Advantages	Disadvantages
Entering data is very fast . You just swipe the card through the machine.	The magnetic stripe on cards can only hold a small amount of data .
No possibility of data entry errors as there is no typing involved.	The cards need to be in contact with the reader in order to be used. (cannot be used over the internet for example)
The stripes on cards are not affected by water and so quite robust (not easily damaged).	If the stripe does get damaged then the data is lost (unreadable).
Data held on the card cannot be read by eye so bank details etc are safe.	Magnetic stripes on cards are easily duplicated (known as card cloning). (if this happens you are prone to theft of funds in your bank)



Hotel room cards contain data on the stripe that will unlock the door once inserted into the reader.

2 Chip and Pin Reader

Information:

- # These allow people to **pay for goods and services electronically** at **EFTPOS** terminals (electronic funds transfer point of sale).
- # The chip and pin reader works by inserting a bank/credit card into a slot and then entering a **PIN** (personal identification number).

The PIN ensures that the person with the bank card is the **genuine owner**. Only the owner of the card should know the PIN and money can only be transferred with the correct number.

- # If the correct PIN is entered, the cost of goods/services will be taken from the card holder's bank and transferred to the companies (restaurant, store etc)

Uses of Chip and Pin Readers:

- # Used to make secure payments for goods/services in places such as:
 - **Supermarkets**
 - **Restaurants**
 - **Cafes**
 - **Buying petrol**
 - **etc**

Examples:



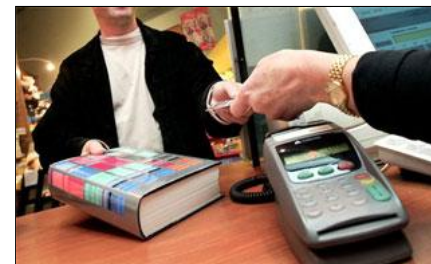
Chip and PIN Readers require the correct identification number to be entered.



Advantages/Disadvantages of using Chip and PIN Readers:

Advantages	Disadvantages
More secure than magnetic stripe readers as the user needs to know the correct PIN.	It is possible to forget your PIN which means you cannot make payments with your card .
Information contained on the chip is harder to clone (copy) than it is to copy magnetic stripes .	People can look over your shoulder as you enter your PIN. This makes it possible to access your money if they ever stole your card.
Chips can hold more data than magnetic stripes can.	
Chip and PIN Readers are portable so they can be brought to you (waiter in a restaurant for example)	

Information on a card's chip is more secure than on a magnetic stripe .



Many goods and services can be purchased using Chip and PIN Readers .

3 Barcode Scanner/Reader

Information:

- # Used to **read information** that is stored on product's **barcodes**.
- # The barcode scanner reads the information stored on the barcode using a **visible red light** which is reflected and translated into digital information.
- # The barcode is present on products and holds information about each product including:
 - **Product Id number**
 - **Manufacturer**
 - **Country of origin**

The barcode **does not store the price of the product**. This information is held in a **computer database** and accessed using the product id.

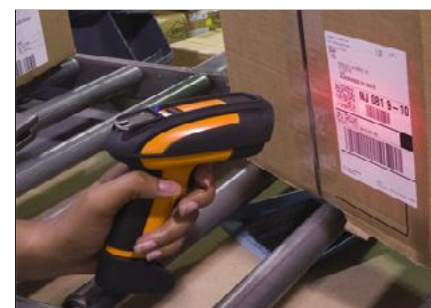
Note: By storing the price in a database instead of on individual bar codes, it is easier to change (if you had a sale for example). All you would do is change the price in the database instead of having to replace the barcodes on each of the items.

- # Once a barcode has been scanned, a **computer can read the information** stored on the barcode and **access details about the product** that are stored in a database.
- # Barcode scanners are usually **built into POS** (point of sale) terminals in supermarkets.
- # **Handheld scanners** are also commonly used to read bar codes (usually if items are too large and heavy to move to the scanner).

Examples:



Barcodes contain information about products and can be read with scanners.



A barcode on a package being scanned.

Uses of Barcode Scanners/Readers:

- # Used in **supermarkets, stores** and **warehouses** where goods are marked with a barcode.

Remember: Barcodes hold **information about the product** which is **linked to a computer database**. This enables automatic very fast itemised billing to take place.

- # They are used in **libraries** to **scan in library cards** and **read ISBN numbers** on books to find out which ones are on loan.
- # They are used in **keeping track of packages** that are being delivered to different locations.
- # Also used in **organising luggage in airports** (helps ensure that luggage is loaded onto the correct plane).

Advantages/Disadvantages of Barcode Scanners/Readers:

Advantages	Disadvantages
Quick and easy to change product prices. Only the price data in the database needs to be updated. There is no need to change individual prices on each of the item's bar codes.	Quite an expensive way to gather information. Each item needs a barcode and the scanners / computer technology / database needed is also expensive.
Quick and easy way of getting information into a system. (all you do is scan the item's bar code)	Bar codes can be swapped on items which means people may pay less for more expensive items.
Very accurate as there is no manual typing involved.	Rely on undamaged bar codes in order to function. (if a bar code is damaged, item numbers have to be typed into the system by hand)
Enables automatic stock control.	

4 Optical Mark Reader (OMR)

Information:

- # System which automatically reads **marks made in pen or pencil**.
- # Typically used to **read multiple choice examinations** where students shade in (mark) answers that they think are correct.



Barcodes are found on items that we buy in stores.



POS terminals in supermarkets have built-in barcode scanners.



Hand held barcode scanners are used to read barcodes on large items.

Examples:

- # The OMR reader **shines a light onto the form** and **less light is reflected** where a pencil mark has been made.
- # This allows the reader to tell which **options have been chosen** and sends the results back to a computer.

Once the data is on a computer it can be **quickly analysed**.

For example: the computer will work out the grade of a multiple choice test.

- # Optimal Mark Readers can input the data held on forms (such as multiple choice exams) **very fast**.

Up to **4000 sheets per hour** can be processed this way.

Imagine how long it would take to mark 4000 multiple choice tests by eye!!



Optical Mark Reader automatically inputting the answers given on a multiple choice form.



OMR devices can read the shaded areas on multiple choice tests.

Uses of OMR's:

- # Used to read the shaded areas on documents such as:

- **Multiple choice examinations**
- **Multiple choice questionnaires**
- **Lottery tickets**

But are typically used on any document that has multiple choice responses that are shaded in.



They can also read shaded areas on lotto tickets.

Advantages/Disadvantages of OMR's:

Advantages	Disadvantages
Extremely fast way of inputting information into a computer. (around 4000 forms can be read per hour)	Forms need to be accurately designed to make sure that the marks/shading are in correct position to be read by the device.
Very accurate as document's data is fed into the computer automatically with no manual typing. (accuracy relies on the forms being filled in correctly)	If forms have been incorrectly filled in they cannot be read properly . (In this case documents would have to be read manually, by eye, which would take a long time)
Can save money as there is no need to employ workers to enter form information into a computer.	Can only read shaded areas (Cannot input or read text).
	Forms can jam in the machine as they are fed through.

5 Optical Character Reader (OCR)

Information:

- # Optical character readers consist of a **scanner** along with **special software**.
- # OCR's allow us to **scan text on paper** (hardcopy) into a computer.
- # Once into the computer, OCR software **converts** the **scanned text** into a **digital format**.

The digital version of the text can be **edited** and **formatted**.

- # Once converted to digital, the text can be used in programs such as **word-processors**, **presentations**, **web sites etc**. See diagram below:

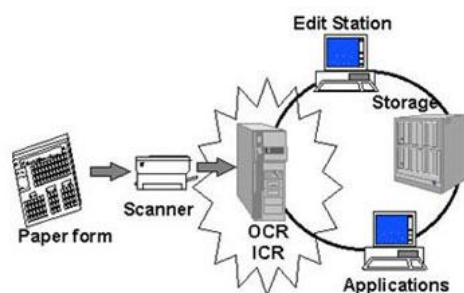


Diagram shows how OCR works:

1. Hard copy scanned into computer
2. OCR software used to convert scanned text into digital (digitizes)
3. Digital text can be used in digital devices and applications

Examples:



Printed text on a document being scanned as part of the 'digitizing' process.



Regular scanners can also be used for inputting scanned text into a computer ready for digitizing.

Uses of OCR's:

- # Used in the **processing of Passports** and **Identity Cards**.
- # OCR is also used in the process of **digitizing books**.

Note: book digitizing is where **hard copy printed books** are **converted into digital format** for use on E-book readers such as Amazon's Kindle.

Advantages/Disadvantages of OCR's:

Advantages	Disadvantages
Very fast way of entering hard-copy text	The system can have difficulty reading



Image of text being converted to digital.
(click image to zoom)

into a computer.	handwriting (especially if it is messy).
Cheaper than paying an employee to manually re-type printed text into a computer.	Not 100% accurate. (relies on being able to read the original text)
Avoids typing errors as the process is automatic (no typing involved)	As it is not 100% accurate, information needs to be manually checked for errors . This is time consuming .



It is also possible to digitize handwriting.

6 Magnetic Ink Character Recognition (MICR)

Information:

- # MICR is a system which can read information printed in a **special ink** (the ink contains iron particles).
- # This special ink is used to write **customer account numbers** on the bottom of **cheques**.
- # The MICR device reads the account numbers then **converts it** to a form the computer can understand.
- # The converted information is then **sent to the computer** so the cheque can be **processed**.

Uses of MICR:

- # MICR devices are mainly used to **process cheques**. Below there is a summary of the process:

1. Every day millions of cheques. are sent to banks for processing. To process the information on these cheques. would take far too long if it were done by hand.
2. MICR automatically (and quickly) processes the information on these cheques. with **100% accuracy** (They can handle 300 per minute).
3. MICR can be used to read the following information:
 - **Customer account number**
 - **Bank sort-code/branch number**
 - **Cheque number.**

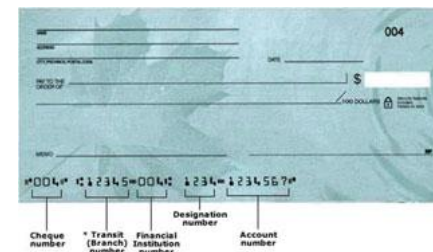
- # Some countries also use MICR devices to read the **passport number on passports**.

Advantages/Disadvantages of MICR:

Examples:



A MICR device reading the magnetic ink on a cheque.



Magnetic ink is used to write various account numbers on cheques. (click image to zoom)

Advantages	Disadvantages
Cheques. can be processed very quickly (up to 300 per minute.)	Only a very limited number of different characters can be read.
Greater security than OCR because the special ink characters cannot be changed.	Both the MICR device Magnetic Ink are more expensive than other methods of direct data entry.
No manual input... therefore less errors . (user just swipes information through the reader)	
Information can still be read even if someone writes over the magnetic ink characters. For example: if someone tries to manually alter an account number by hand the device will still read only the original data.	



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